

Cascais Race Brief — Swedish Team

J/70 World Championship, Cascais · debrief of Tuesday 8 September, and wind outlook to Friday 11 September

Race area **38.68 N 9.42 W** Fleet **34 boats** Telemetry held **14 boats** Model bias **+18.6°** Tacking angle **80.5°**

The one number that changed

The GRIB sits **+18.6° to the right** of the wind the fleet actually raced — not the +11° carried in yesterday's version. The earlier figure averaged the committee's line square with the fleet's beat bearing, which assumes a square line. The event reports show the line was 6° then 12° off square, so that estimate was dragged toward the model. Measured error was **+19.1°** in race 1 and **+18.0°** in race 2 — tighter than the old estimate, which is what a real local bend looks like. **Working rule: take the GRIB and subtract 18–19°** whenever it is a 10–14 kn gradient Nortada.

Start line — the day's biggest single factor

Race	Line	Bias	Advantage	Favoured	Cross-check
R1	1001 m	6°	107 m	Pin	105 m
R2	1007 m	12°	206 m	Pin	209 m

Pin favoured in both races. Race 2's 206 m on a 1007 m line is a fifth of the line's length — roughly 17 boat lengths handed out before anyone had sailed a metre. Our own formula, given the same line and bias, returns 105 m and 209 m: within 2% of the event's figures. The telemetry says the same thing independently — boats starting nearer the committee boat took longer up the first beat in both races (r = 0.55 and 0.36).

Where every boat started

Distance to line at the gun, metres (positive = behind). Position along the line: 0.00 = pin, 1.00 = committee boat.

Race	Boat	To line	Along	Third
R1	J70 Let it be	+2.2	0.89	boat
R1	noticia	+4.3	0.91	boat
R1	Dime Piece	+4.3	0.98	boat
R1	Florida 65	+4.8	0.44	middle
R1	joust70	+4.9	0.87	boat
R1	Team Sweden (Roman)	+5.2	0.45	middle
R1	GTG New 2026	+5.8	0.65	middle
R1	Lady in red 2	+5.9	0.36	middle
R1	Patakin 3	+6.5	0.28	pin
R1	Ba ba	+7.4	0.59	middle
R1	Phantom7	+13.0	0.91	boat
R1	Mag	+14.0	0.71	boat
R1	Yupi	+17.2	0.94	boat
R2	Lady in red 2	+3.1	0.30	pin
R2	noticia	+3.5	0.26	pin
R2	joust70	+4.3	0.13	pin

Race	Boat	To line	Along	Third
R2	Ba ba	+5.2	0.47	middle
R2	Yupi	+6.7	0.36	middle
R2	J70 Let it be	+10.8	0.91	boat
R2	Dime Piece	+10.9	0.51	middle
R2	Team Sweden (Roman)	+11.1	0.93	boat
R2	Patakin 3	+13.6	0.08	pin
R2	Florida 65	+13.7	0.38	middle
R2	Mag	+15.5	0.09	pin
R2	Phantom7	+18.3	0.35	middle

Wind measured on the water, by leg

Race	Setting	1st beat	1st run	2nd beat	2nd run
R1	318°	317°	323°	313°	317°
R2	313°	314°	321°	313°	313°

The runs read consistently right of the beats — 323° against 317° in race 1, 321° against 314° in race 2. Same sign both races, so it is worth treating as real: expect the breeze a few degrees right once you are heading down.

Race 1 — legs and roundings

Mark times in minutes from the start. Extra distance = how much further than the direct line each boat sailed up the first beat.

#	Boat	M1	M2	M3	Beat kn	Run kn	Extra m	Tacks
1	Patakin 3	32.0	48.8	74.8	5.92	12.16	1134	0
2	Team Sweden (Roman)	32.7	49.5	75.8	5.88	11.67	1249	2
3	GTG New 2026	32.5	50.0	76.4	5.98	11.70	1300	1
4	Lady in red 2	33.5	50.3	77.0	5.84	11.44	1324	2
5	noticia	33.3	50.3	77.2	5.91	11.58	1367	2
6	Ba ba	32.7	49.9	77.4	5.93	11.83	1268	2
7	Phantom7	34.3	50.5	77.7	5.95	11.73	1507	7
8	joust70	33.4	51.1	77.8	5.83	11.82	1250	1
9	Yupi	33.7	51.4	78.0	6.04	11.57	1460	3
10	Florida 65	32.9	51.4	78.4	5.96	11.52	1345	0
11	J70 Let it be	32.9	51.3	78.7	5.97	10.86	1330	0
12	Mag	34.0	52.5	78.8	6.08	10.99	1596	2
13	Dime Piece	33.1	51.3	79.0	6.11	11.37	1468	0

Us in race 1: 3rd at the first mark, up to 2nd and held it — 1.01 min off the leader at the last rounding. Beat 1 speed was -0.08 kn against the fleet but we sailed **-114 m less distance** than the fleet average, on the fleet-median tack count. That is a well-sailed beat: same speed, shorter track.

Race 2 — legs and roundings

#	Boat	M1	M2	M3	Beat kn	Run kn	Extra m	Tacks
1	Ba ba	33.0	48.8	75.9	5.75	12.34	1198	0
2	Yupi	33.4	49.8	77.2	5.66	11.74	1110	2
3	joust70	34.5	50.8	77.4	5.66	12.18	1340	3
4	Patakin 3	33.9	50.5	77.7	5.61	11.65	1168	1
5	noticia	33.3	50.1	78.0	5.77	11.82	1225	0
6	Mag	34.9	51.1	78.3	5.63	12.11	1361	2
7	Phantom7	35.0	51.5	78.8	5.77	12.10	1495	4

#	Boat	M1	M2	M3	Beat kn	Run kn	Extra m	Tacks
8	Florida 65	35.1	51.2	79.1	5.76	12.29	1545	2
9	Lady in red 2	34.5	51.2	79.3	5.65	11.94	1346	2
10	J70 Let it be	35.3	51.8	79.9	5.82	11.26	1635	1
11	Dime Piece	34.2	51.3	80.0	5.79	11.66	1435	2
12	Team Sweden (Roman)	35.1	52.4	80.6	5.70	11.81	1491	6

Us in race 2: 11th, 12th, 12th — 4.70 min down. Boat speed was fine (-0.02 kn vs fleet). The cost was the track: **+4 tacks above the fleet median** and **+140 m of extra distance** up the first beat, having started 0.93 along the line — the committee-boat end, on a day the pin paid 206 m. The start put us on the wrong side and the beat was spent recovering.

Wind outlook to Friday

Day	Speed	Gust	Corrected direction	Shift	Model spread
Wed 9 Sep (today)	11–14 kn	30 kn	322° → 319°	-3°	7°
Thu 10 Sep	5–12 kn	24 kn	338° → 320°	-18°	8°
Fri 11 Sep	2–8 kn	18 kn	43° → 317°	-87°	110°

Today is the breeze day. 11–14 kn building through the afternoon, gusts to 30, direction steady around 316–322° corrected. Textbook Nortada and the models agree to within 7°.

Thursday starts soft. 5 kn at midday, only filling to 11–12 kn by late afternoon, with the direction swinging 338° → 320° as the sea breeze takes over from the gradient. Expect a postponement and a big left shift as it fills.

Friday looks marginal. 2–3 kn until mid-afternoon, 8 kn at best by 18:00. The models disagree by up to 110° at midday, which is what no gradient looks like — direction is not meaningful at that point. **The 18–19° correction should not be applied below about 8 kn:** it was calibrated on a gradient Nortada, and a thermally-driven light day bends differently.

Hour by hour, racing window

Day	Local	Raw GRIB	Corrected	kn	Gust	Spread
Wed 9 Sep	12:00	341°	322°	11.2	24	2°
Wed 9 Sep	13:00	339°	320°	11.1	24	3°
Wed 9 Sep	14:00	336°	317°	11.8	26	6°
Wed 9 Sep	15:00	335°	316°	12.5	28	7°
Wed 9 Sep	16:00	335°	317°	13.1	29	2°
Wed 9 Sep	17:00	336°	317°	13.3	30	0°
Wed 9 Sep	18:00	337°	319°	13.5	30	2°
Thu 10 Sep	12:00	357°	338°	5.1	15	8°
Thu 10 Sep	13:00	350°	331°	5.0	15	7°
Thu 10 Sep	14:00	338°	320°	6.0	17	6°
Thu 10 Sep	15:00	333°	315°	8.6	20	3°
Thu 10 Sep	16:00	334°	316°	11.0	24	4°
Thu 10 Sep	17:00	336°	317°	10.9	24	4°
Thu 10 Sep	18:00	339°	320°	11.5	24	3°
Fri 11 Sep	12:00	62°	43°	1.5	11	83°
Fri 11 Sep	13:00	330°	311°	1.9	12	110°
Fri 11 Sep	14:00	340°	322°	2.7	12	74°
Fri 11 Sep	15:00	328°	310°	4.3	13	34°
Fri 11 Sep	16:00	330°	312°	6.0	15	20°
Fri 11 Sep	17:00	332°	314°	7.4	18	13°
Fri 11 Sep	18:00	335°	317°	8.0	18	6°

What is not in this yet

- **Per-leg VMG, best/worst moments, places gained per leg.** The brief's priority. My own leg-splitting from the raw logs is not yet reliable — it produced identical leg lengths across all boats, which is the detector failing, not the boats agreeing. The leg tables above come from the validated mark detections instead. Fixing this is the first job for the 18:00 model.
- **Offsets.** Four legs and two offsets per race — the offsets are not separated out here.
- **Official start list and results.** Not yet pulled; positions above are order at the last rounding, which is not finishing order. The final leg is excluded because the loggers' RACE_END is synced across the fleet and so is not a finish signal.
- **Course and mark GPS.** Mark positions are inferred from track reversals, not recorded, so extra-distance figures carry unknown error. The committee's actual mark coordinates would fix this.
- **20 of the 34 boats.** Every fleet-relative figure is against a 14-boat sample. **MidlifeCrisis** logs at ~10 Hz — the best file in the fleet — and is not in any analysis yet; **ToNessa** is in the GitHub repo but never reached the Drive folder.
- **A start time that does not reconcile.** The event report puts race 1 at 13:34:52 local; every logger recorded its timer expiring at 13:05:00, all within ~20 ms of each other. Synced to each other, not to the gun. Treat any "at the gun" number as approximate until this is settled.
- **Wind on the boat.** No boat logs a wind instrument, so every measured bearing here comes from GPS geometry or the event's own analysis. TUR 442 and TYRA already feed NMEA speed, depth and temperature into the Atlas — adding a masthead feed would remove the need for the bias correction entirely.

Vakaros Atlas VKX telemetry decoded at 0.5 s from the Sailing drive folder, 8 September 2026, 14 of 34 boats · measured wind and start-line figures from the event race reports (ib-sailing.com) · forecast from a live ICON-EU / ECMWF IFS / GFS consensus at 38.68 N 9.42 W, corrected by -18.6° · prepared 09 Sep 2026 10:14 local.